Kapil Chawla, Ph.D.

Computational and Applied Mathematics

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Google Scholar GitHub

Research Profile

I am a Postdoctoral Research Associate in the Department of Mathematics at Florida State University, where I work on a U.S. Department of Energy-funded project on multiphysics machine learning for subsurface energy systems. My research combines numerical methods for partial differential equations with scientific machine learning, focusing on nonsmooth solutions, discontinuous coefficients, and heterogeneous multiphysics systems. During my Ph.D., I studied stationary and time-dependent obstacle formulations for ice-sheet dynamics and developed feedforward and physics-informed neural solvers. This work led to DG-FEONet, a data-free neural operator based on SIPG discretization. My current research develops sparse reduced-order models and mesh-free solvers for geothermal and porous-media systems. I also study hybrid quantum-classical finite element solvers for heterogeneous Darcy flow.

Research Expertise

Scientific Machine Learning & Neural Operators • Obstacle Problems and Variational Inequalities • Reduced-Order Modeling & Dimensionality Reduction • Computational PDEs • Finite Element Methods • Quantum Computing • Multiphysics & Subsurface Energy Systems

Employment

2025–Present **Postdoctoral Research Associate**, Department of Mathematics, Florida State University, Tallahassee, Florida, USA.
DOE Project: *Advanced Multi-Physics Machine Learning for Subsurface Energy Systems Across Scales*, Grant No. DE-SC0024703.

2019–2025 **Associate Instructor**, Department of Mathematics, Indiana University Bloomington, Bloomington, Indiana, USA.

2018–2019 **Assistant Professor**, Sri Guru Granth Sahib World University, Fatehgarh Sahib, India.

2016–2018 **Junior Research Fellow**, Scientific Analysis Group, DRDO, Delhi, India.

2015–2016 **Assistant Professor**, S.D. College, Ambala Cantt, India.

Education

2019–2025 **Ph.D. in Mathematics**, Indiana University Bloomington, Bloomington, Indiana, USA.

2013–2015 **M.Sc. in Mathematics and Computing**, Thapar University, Patiala, India.

2010–2013 **B.Sc. (Non-Medical with Computer Application)**, Punjabi University, Patiala, India.

Publications

1. **K. Chawla**, Y. Hong, J. Lee, and S. Lee, "Discontinuous Galerkin Finite Element Operator Network for Solving Non-smooth PDEs," *SIAM Journal on Scientific Computing (SISC)*, accepted, 2026. **arXiv** • **Code**.
2. **K. Chawla** and W. Holmes, "A Physics-Informed Neural Network Approach to Simulating Ice Dynamics Governed by the Shallow Ice Approximation," *International Journal of Numerical Analysis and Modeling*, vol. 23, no. 4, pp. 523–550, 2026. **Journal** • **arXiv**.
3. **K. Chawla**, S. Lee, and Y. Shin, "PAM: Parallel and Adaptive Mesh-Free Methods for Flow in Heterogeneous Porous Media," submitted to *Journal of Computational Physics*, 2026. **arXiv**.
4. **K. Chawla**, S. Lee, and J. Valyou, "A Hybrid Quantum-Classical Coarse-to-Fine Finite Element Framework for Heterogeneous Darcy Flow," under revision in *SIAM Journal on Scientific Computing (SISC)*, 2026.
5. **K. Chawla**, T. Koike, J. Bessac, and S. Lee, "SparseModesNet for coupled thermal-hydraulic-mechanical system in heterogeneous porous media," submitted to Computational Mechanics, 2026.
6. **K. Chawla**, W. Holmes, and R. Temam, "From Obstacle Problems to Neural Insights: Feed Forward Neural Network Modeling of Ice Thickness," *International Journal of Numerical Analysis and Modeling*, vol. 22, pp. 1–20, 2025. **Journal** • **arXiv**.
7. **K. Chawla** and J. Kaur, "Common Fixed Point Results for Weakly Isotone Increasing Mappings in Partially Ordered b-Metric Like Spaces," *International Journal of Functional Analysis, Operator Theory and Applications*, vol. 7, no. 1, pp. 19–56, 2015.

Manuscripts in Preparation

1. **K. Chawla**, S. Lee and Y. Hong, "Finite Element Operator Network for Obstacle Problems," manuscript in preparation, 2026.

Invited and Selected Presentations

2027 — **Invited Speaker**, Joint Mathematics Meetings (JMM 2027), Chicago, Illinois, USA, January 12–15, 2027.

2026 — **A Multiscale Operator Network Framework for Coupled THM Processes in Geothermal Systems**, invited minisymposium presentation, SIAM Conference on Mathematics of Data Science (MDS26), Salt Lake City, Utah, USA.

2026 — **SparseModesNet: Sparse Nonlinear Dimensionality Reduction for Thermo-Poroelastic Systems**, oral presentation, 2026 Fall Southeastern Sectional Meeting of the American Mathematical Society (AMS), Kennesaw State University, Kennesaw, Georgia, USA.

2026 — **DG-FEONet: Discontinuous Galerkin Finite Element Operator Network**, oral presentation, Spring 2026 Finite Element Circus, Tufts University, Medford, Massachusetts, USA.

2026 — **DG-FEONet: A Data-Free Neural Operator for PDEs with Discontinuous Coefficients**, poster presentation, 4th Southeast Applied & Computational Math Student Workshop, Emory University, Atlanta, Georgia, USA.

2025 — **DG-FEONet: Learning Discontinuous PDE Solutions with Neural Networks**, 12th Graduate Student Conference, Clemson University, Clemson, South Carolina, USA.

2025 — **Modeling Ice Sheet Dynamics Using Physics-Informed Neural Networks: From Stationary to Time-Dependent Frameworks**, ACM Seminar, Florida State University, Tallahassee, Florida, USA.

Technical Skills

Programming & Scientific Computing: Python, PyTorch, FEniCS, NumPy/SciPy

Teaching Experience

2021–2024 **Instructor of Record**, Indiana University Bloomington; Math-M125: Precalculus, Math-M127: Precalculus and Trigonometry, Math-M118: Finite Mathematics.

Fall 2024 **Course Coordinator Assistant**, Indiana University Bloomington; Math-M125: Precalculus.

2019–2021 **Recitation, Instructional Support, and Grading**, Indiana University Bloomington; Calculus, Applied Calculus, Differential Equations, and Elementary Computational Methods.

2018–2019 **Assistant Professor (Instructor of Record)**, Sri Guru Granth Sahib World University; Numerical Methods, Programming in C, Fundamentals of Computer Science, and Real Analysis.

2015–2016 **Assistant Professor (Instructor of Record)**, S.D. College; Calculus, Advanced Calculus, Numerical Analysis, Real Analysis, Measure and Integration, Algebra & Number Theory, and Programming Skills with C.

Guest Lectures

Apr 14, 2026 **Guest Lecture on Physics-Informed Neural Networks (PINNs)**, Florida State University, Tallahassee, Florida, USA.

Apr 16, 2026 **Guest Lecture on Operator Networks**, Florida State University, Tallahassee, Florida, USA.

Student Mentoring

Fall 2024 **Graduate Mentor**, *Solving Parabolic Obstacle Problem and Ice Sheet Model Predictions through Deep Learning*; mentored Joseph, Noah, Christopher, and Fredrich.

Spring 2024 **Graduate Mentor**, *Enhancing Ice Thickness Predictions of the Greenland Ice Sheet through Deep Learning*; mentored Zake, Will Amos, and Elijah.

Fall 2023 **Graduate Mentor**, *Numerical Solutions to Poisson and Heat Equations: A Finite Difference Approach*; mentored Tommy Spencer Pugh.

Awards and Honors

2024 **Travel Award**, Indiana University Bloomington.

2023 **Travel Award**, Indiana University Bloomington.

Dec 2016 **CSIR UGC NET (Mathematical Sciences)**, All India Rank 21.

Jun 2016 **CSIR UGC NET (Mathematical Sciences)**, All India Rank 14.

Professional Development

2023 **SLMath-IBM Summer School**, IBM Research, Almaden, California, USA.

2022 **Graduate Student Mathematical Modeling Camp**, University of Delaware, Newark, Delaware, USA.